

USE CASE 2: AI for Backend API Scaffolding (Contract-First)

Covers:

- AI for Backend
 - Code generation patterns
 - API scaffolding prompts
 - Structured outputs
 - Validation & retries
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Scenario:

The Product and Cart services need REST APIs designed using OpenAPI-first approach.

The AI Must:

- Generate OpenAPI spec
 - Generate controller scaffold
 - Follow error model
 - Maintain backward compatibility
 - Retry if schema validation fails
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Example Prompt:

"You are a backend engineer.
Generate OpenAPI 3.0 spec for Product Service.
Endpoints:

- GET /products
- GET /products/{id}
- POST /products

Constraints:

- Include proper HTTP status codes
- Structured error model

- Versioning: v1
 - Output strictly in OpenAPI YAML format."
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What to Build:

- OpenAPI YAML
 - Backend scaffold code
 - Validation layer
 - Retry vs re-prompt demonstration
 - Max token limit experiment
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Deliverables:

- OpenAPI file
- Backend scaffold
- Validation logs
- Prompt refinement documentation